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RESEARCH ARTICLE

Toward Personalized Adaptive Learning Using Artificial Intelligence and Physiological Signals

JUNGMIN HWANG AND CHEOLKYU SHIN, (Member, IEEE)
Division of AI Convergence, Hallym University, Chuncheon 24252, South Korea

Corresponding author: Cheolkyu Shin (cheolkyu@hallym.ac.kr)

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ABSTRACT Personalized Adaptive Learning (PAL) systems aim to dynamically tailor instruction to learners' cognitive and affective states, yet most existing approaches lack real-time, scalable indicators of learner status. Advances in wearable sensing and Internet of Things (IoT) technologies now enable unobtrusive monitoring of signals that reflect cognitive load, fatigue, and arousal, providing a foundation for physiology-informed PAL. This paper presents a PAL system that integrates real-time physiological deviations and accumulated fatigue into a performance update model, enabling session-wise modeling of learner performance. Based on this performance update model, we construct multiple difficulty adaptation policies. These policies range from rule-based heuristics to Artificial Intelligence (AI)-based strategies that optimize the PAL system. The AI-based methods leverage pseudo labels derived from physiological indicators and modeled performance dynamics, allowing supervised learning of adaptation policies without ground-truth performance data. Using the publicly available Psychological State Identification (PSI) dataset, we simulate multi-session learning scenarios and demonstrate that AI-based policies rapidly converge to near-perfect performance, consistently outperforming rule-based baselines. We also discuss key deployment challenges, including signal reliability, explainability, and the need for standardized learning analytics frameworks in real-world PAL systems.

INDEX TERMS Personalized adaptive learning (PAL), physiological signals, artificial intelligence (AI), wearable sensing devices, Internet of Things (IoT).

I. INTRODUCTION

With rapid advances in Artificial Intelligence (AI), the focus of Educational Technology (EdTech) is increasingly shifting toward the realization of Personalized Adaptive Learning (PAL) [1], [2], [3], [4]. To enable effective PAL, it is essential to collect diverse learning-related data and analyze learners' states using AI-driven methods [5], [6], [7]. Moreover, there is a growing need for research and standardization to define the policies and protocols that govern when and how AI systems should deliver PAL interventions [8], [9], [10], [11], [12].

Among the various types of data used to understand learner status—such as behavioral logs, response times, and facial expressions—physiological signals such as Heart Rate Variability (HRV) and Galvanic Skin Response (GSR) offer direct, real-time measurements of a learner's internal state [13], [14]. In authentic educational environments, the data acquisition process should be both seamless and feasible. Recent advances in Internet of Things (IoT) infrastructures and wearable technologies—such as smartwatches, biosensors, and ambient sensing devices—have made the real-time collection of physiological data increasingly viable [15], [16], [17], [18]. However, despite the availability of such data, current EdTech systems rarely incorporate physiological signals into data-driven PAL. In most classrooms, instructors still rely on traditional teaching methods without real-time insight into learners' cognitive or emotional states. Consequently, instructional decisions are often applied uniformly across learners, overlooking the dynamic and individualized needs that arise during the learning process.

In addition, recent advances in AI—particularly in natural language processing, learning analytics, and personalized recommendation—have enabled the development of

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AI-driven EdTech systems. However, practical solutions that effectively address the needs of both instructors and learners remain limited. To bridge this gap, future AI-driven EdTech systems should integrate continuous physiological monitoring with intelligent decision-making mechanisms. Such integration is essential for realizing PAL systems that dynamically tailor instruction based on a learner's moment-to-moment state—thereby maximizing engagement, cognitive alignment, and ultimately, learning outcomes [19], [20].

A. RELATED WORK

Previous studies have investigated how physiological signals can be leveraged to monitor cognitive and affective states, laying the groundwork for their potential integration into PAL frameworks [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. These studies demonstrate the potential of physiological signals to detect cognitive load, model emotional states, and trigger timely instructional interventions tailored to individual learners. Early research primarily focused on single-modality[1] physiological signals. For instance, Antonenko et al. [21] demonstrated that electroencephalography (EEG) can serve as a continuous and online measure of cognitive load, highlighting the sensitivity of alpha-band activity to mental workload. Berka et al. [22] validated real-time EEG-based engagement and workload indices in realistic task environments, showing strong correlations with task difficulty and performance. Solhjoo et al. [23] further confirmed that Heart Rate (HR) and HRV dynamically reflect cognitive load in clinical reasoning tasks.

Building on these foundations, more recent work has advanced toward multimodal[2] sensing approaches that combine multiple physiological and behavioral signals. Woolf et al. [24] incorporated biosensors into intelligent tutoring systems to detect affective states such as frustration and boredom. Kapoor and Picard [25] showed that combining facial expressions, posture, and GSR enables accurate, real-time detection of frustration and interest. A comprehensive meta-analysis by D'Mello and Kory [26] confirmed that multimodal systems—especially those incorporating physiological signals—consistently outperform unimodal systems in affect detection. Monkaresi et al. [27] demonstrated that combining HR with facial expression features enhances engagement classification. In applied settings, Begum et al. [28] classified wheel loader operators' states using Multivariate Multiscale Entropy combined with case-based reasoning (MMSE–CBR), while Chen et al. [29] analyzed multimodal physiological features with kernel-based classifiers to identify driving stress. More recently, Abbad-Andaloussi et al. [30] examined declarative process model complexity and assessed cognitive load using multimodal data, further demonstrating the potential of heterogeneous signals in capturing mental effort.

A recent trend extends these works by integrating physiological signals with AI-based adaptation [31], [32], [33], [34], [35], [36], [37], [38], [39]. These systems treat physiological signals as dynamic input to machine learning models that drive real-time instructional decisions. For example, Standen et al. [31] developed a tutor for students with intellectual disabilities that adjusted content delivery based on affective states inferred from GSR and HR. Xiong et al. [32] proposed a cognitive load recognition method using EEG and electrocardiography (ECG) signals, demonstrating that multimodal physiological features combined with machine learning algorithms can effectively classify different levels of cognitive load. Wu et al. [33] introduced a deep learning model that classifies emotions using fused EEG and eye-tracking data. Nasri [34] designed a physiological adaptation framework in virtual reality (VR) training, using HRV and eye-tracking data with a machine learning model to estimate cognitive load and stress, dynamically adjusting task difficulty. Wang [35] introduced a fatigue-aware adaptive interface that uses HR and gaze features, processed through a deep neural network and reinforcement learning, to modify User Interface (UI) elements such as font size and color contrast in digital reading environments. In the educational context, Rehman et al. [36] measured student attention using EEG signals and deep reinforcement learning, illustrating how advanced AI techniques can leverage physiological signals to support adaptive instruction. Finally, recent studies [37], [38], [39] have explored the use of Large Language Models (LLMs) to analyze learners' emotional, behavioral, and physiological states. These works highlight the growing potential of LLMs for personalized learning, emotion recognition, and health monitoring.

Prior studies have demonstrated that physiological signals can be used to infer cognitive load, affective states, attention, and fatigue, and that AI-based models show promise in supporting real-time adaptive instruction. However, despite these advances, the quantitative relationship between physiological deviations and learning performance remains insufficiently understood, and only limited empirical evidence explains how such deviations translate into measurable performance change. Moreover, existing single and multimodal AI-driven approaches have largely overlooked the need to model physiological deviations, accumulated fatigue, and adaptive policy learning within a unified and principled PAL framework. To address these gaps, the present work provides an initial theoretical and simulation-based foundation that integrates these elements and offers a structured perspective on how physiological indicators may inform adaptive learning strategies.

B. MOTIVATION AND CONTRIBUTIONS

Recognizing the importance of PAL using AI and physiological signals from the related work, the following

[1] Single-modality refers to approaches that rely on physiological signals from a single physiological system or sensing modality (e.g., cardiovascular signals such as HRV and skin temperature).

[2] Multimodal approaches integrate signals from multiple distinct systems (e.g., EEG and GSR, or HR and facial expressions) to improve robustness and accuracy in detecting cognitive or affective states.

questions need to be addressed: (1) How can learners' cognitive load be reliably and continuously detected in real-world learning environments using physiological signals in a non-intrusive and scalable manner? (2) What types of difficulty adaptation policies can be effectively implemented in PAL systems based on physiological feedback, and how can these policies dynamically respond to fluctuations in learners' physiological states? (3) What practical challenges must be addressed to enable the deployment of AI-driven, physiology-informed PAL systems in authentic classroom settings?

Addressing these three questions constitutes the central motivation of this work. To begin with the first question, we consider the feasibility of capturing physiological feedback in real-world learning environments using wearable devices (e.g., smartwatches)[3] connected through an IoT infrastructure. In particular, we utilize the Psychological State Identification (PSI) dataset [40], which includes physiological data commonly available via smartwatch sensors and annotated with learners' cognitive load levels. To address the second question, we propose both rule-based and AI-driven difficulty adaptation policies. These policies dynamically adjust instructional difficulty in response to the learner's real-time physiological state, with the aim of optimizing learning performance across multiple sessions. The rule-based policy employs heuristic thresholds, while the AI-driven policy leverages current learner data to personalize adaptation over time. In response to the third question, we identify several critical challenges that must be addressed to enable the deployment of AI-driven, physiology-informed PAL systems in real-world learning environments. These include ensuring the accessibility and reliability of physiological data, addressing privacy and ethical concerns, improving the explainability of AI-driven decisions, and promoting interoperability through standardization. Such challenges are not unique to education; similar issues have been widely documented in other AI-driven physiological sensing domains—including biomedical signal analysis, wearable health monitoring, and AI-assisted medical decision systems [42], [43], [44], [45], [46]. We discuss concrete directions for addressing these challenges and moving toward scalable, trustworthy implementations of PAL in practice.

Based on the motivation above, this paper aims to explore how physiological signals can be utilized for PAL and how intelligent difficulty adaptation policies can enhance learning performance in response to individual physiological states. The key contributions of this work are as follows:

- We address a critical research gap in physiology-informed PAL by formulating an initial quantitative framework that links physiological deviations to learning performance. Our mathematical formulation and simulation-driven analysis establish an early foundation for understanding how physiological indicators may inform adaptive learning strategies.
- We propose a PAL framework that includes a performance update equation incorporating both real-time physiological deviations and accumulated fatigue, enabling session-wise modeling of learner performance;
- We design multiple difficulty adaptation policies, ranging from rule-based heuristics (e.g., Yerkes–Dodson-based [41] and distance-based) to AI-based strategies (e.g., Random Forest [47] and XGBoost [48]), applicable to various physiological input modalities. The AI-based methods generate pseudo labels from physiological indicators and modeled performance dynamics, allowing supervised learning of adaptation policies even in the absence of ground-truth performance data;
- We empirically validate the proposed adaptation policies using the publicly available PSI dataset [40]. The results show that AI-based policies rapidly converge to near-perfect performance (up to 97.6 ± 0.9) with substantial learning gains (+80.5%), consistently outperforming rule-based methods;
- We provide a forward-looking discussion on real-world PAL deployment challenges, including the accessibility and reliability of physiological signals, the explainability of PAL, and the need for standardized learning analytics.

[3] While various physiological sensing modalities exist (e.g., EEG caps, dedicated GSR electrodes), we focus on signals obtainable from wearable devices that are unobtrusive and suitable for continuous, everyday use—such as smartwatches equipped with photoplethysmography (PPG) sensors.

C. PAPER ORGANIZATION AND NOTATION

The rest of the paper is organized as follows. Section II introduces the set of physiological input modalities considered in this work and elaborates on their role in supporting PAL. In Section III, we present the overall system model for PAL, including the mathematical formulation of performance updates, as well as models for fatigue accumulation and physiological signal dynamics across learning sessions. In Section IV, we propose both rule-based and AI-driven difficulty adaptation policies. Section V provides experimental results based on a publicly available PSI dataset, comparing the effectiveness of different policies in improving learning performance. Discussions are provided in Section VI about critical challenges and future directions for deploying PAL in practice, and conclusions are drawn in the last section.

Notations: The notation $\mathrm{clip}_{a,b}(x)$ denotes a clipping function that restricts x to lie within the range $[a, b]$. Specifically, if $x < a$, it is set to a, and if $x > b$, it is set to b. The symbol $\odot$ denotes the Hadamard (element-wise) product. The notation $\|\cdot\|_1$ refers to the L1 norm, defined as the sum of absolute values of vector components. The symbol $\mathcal{N}(0, \sigma^2)$ denotes a normal distribution with zero mean and variance σ^2. The superscript $[\cdot]^\top$ represents the transpose of its argument.

II. VARIOUS PHYSIOLOGICAL INPUT MODALITIES

This section describes the physiological signals included in the PSI dataset [40], focusing on their physiological significance, measurement methods, typical healthy ranges,

TABLE 1. Summary of physiological signals in the PSI Dataset [40].

Signal	Domain	Unit	Typical Range*	PSI Stats**	Sensor Type(s)
HRV	Autonomic Cardiovascular	ms	25–100 (SDNN); 20–70 (RMSSD)	17.4–78.8 (49.9 $\pm$ 9.9)	ECG, PPG
GSR	Autonomic	μS	2–20	0.5–1.5 (1.0 $\pm$ 0.3)	EDA
EEG bands	Neurological	Hz	delta: 0.5–4 alpha: 8–13 beta: 13–30	delta: 0.7–2.1 (1.4 $\pm$ 0.7) alpha: 0.7–2.1 (1.4 $\pm$ 0.7) beta: 0.7–2.1 (1.4 $\pm$ 0.7)	EEG cap
BP	Cardiovascular	mmHg	110/70–130/90 (sys/dia)	110/70–129/89 (119.4/79.4 $\pm$ 5.9/5.8)	Cuff, CNIBP
SpO_2	Cardio-pulmonary	%	95–100	95–100 (97.4 $\pm$ 1.4)	PPG
HR	Autonomic Cardiovascular	bpm	60–100	60–99 (78.8 $\pm$ 11.8)	ECG, PPG
RR	Autonomic Respiratory	bpm	12–20	12–24 (18.1 $\pm$ 3.7)	Belt, PPG
Skin Temp.	Autonomic Peripheral	°C	33–36.5	34–36.0 (35.0 $\pm$ 0.6)	IR therm., Thermistor

* Typical Range: Derived from references presented in Section II-A.
** PSI Stats: Range (Mean $\pm$ Std) calculated from the PSI dataset.
Sensor descriptions: ECG: Electrical heart signal sensor; available as chest straps, patches, or smartwatch-integrated modules. **PPG**: Optical sensor embedded in wrist-worn devices for HR, HRV, SpO_2, and RR. **EDA**: Electrodermal activity sensor, typically via GSR electrodes on fingers or palms; available in wristbands. **EEG cap**: Electrode-based device, typically used in lab environments; limited wearability. **Cuff**: Oscillometric BP monitor for clinical/home use; non-wearable. **CNIBP**: Cuff-free device for continuous BP monitoring; wearable in controlled settings, but impractical for casual use. **Belt**: Chest-worn band to measure respiration; not ideal for continuous use in classroom due to bulkiness. **IR therm.**: Non-contact skin thermometer; not easily wearable. **Thermistor**: Wearable skin temperature sensor embedded in devices like smartwatches or biosensor patches.

and relevance for PAL. These signals span autonomic, cardiovascular, and neurological domains, enabling real-time inference of cognitive and emotional states during learning.

A. PHYSIOLOGICAL SIGNAL CHARACTERISTICS

- **HRV** [ms]: HRV is a key indicator of autonomic and cardiovascular regulation. It reflects the balance and adaptability of autonomic control by quantifying variations in the intervals between successive heartbeats. Accordingly, HRV belongs to the autonomic/cardiovascular physiological domain. HRV is typically derived from an ECG or photoplethysmography (PPG) sensor using metrics such as the standard deviation of normal-to-normal intervals (SDNN)[4] or the root mean square of successive differences (RMSSD)[5] [49], [50]. A lower HRV during challenging tasks implies that the learner is under higher mental strain or stress, whereas higher HRV is associated with better focus and resilience [51].
- **GSR** [μS]: GSR reflects activity of the sympathetic nervous system and belongs to the autonomic physiological domain. It quantifies changes in skin conductance resulting from sweat gland activation. GSR is typically measured using an electrodermal activity (EDA) sensor with electrodes placed on the fingers or palm [52]. This signal captures moment-to-moment variations in arousal and has been widely used to assess cognitive engagement and stress [53]. For instance, sudden spikes in GSR during challenging tasks may indicate heightened arousal, suggesting that the learning content is cognitively demanding or emotionally provocative.
- **EEG bands** [delta, alpha, beta; defined in Hz]: EEG captures electrical brain activity via scalp-mounted electrodes and is classified within the neurological physiological domain. It is typically measured using a scalp-mounted cap configured according to the international 10–20 system [54], [55]. Different frequency bands are associated with distinct mental states: beta activity reflects attention and cognitive processing, alpha is linked to relaxation, and delta indicates drowsiness or fatigue [56]. Analyzing the ratios or temporal dynamics among these bands provides insight into the learner's cognitive state. For example, increased beta or suppressed alpha may indicate heightened focus, whereas rising alpha or emerging delta activity may signal disengagement or fatigue. A high beta-to-alpha ratio, in particular, may reflect elevated cognitive workload or stress—prompting the system to reduce task complexity or insert rest intervals accordingly.
- **Blood pressure (BP)** [mmHg]: BP indicates cardiovascular workload and vascular tone, and is typically measured using either an upper-arm cuff or continuous non-invasive blood pressure (CNIBP) devices. Elevated BP has been associated with increased stress or sustained cognitive effort [57]. An acute rise in BP during a learning task may suggest cognitive overload or frustration, prompting the system to reduce task difficulty or recommend a short break. Sustained moderate increases in BP may also be indicative of accumulated fatigue.[6]
- **Oxygen saturation (SpO_2)** [%]: SpO_2 refers to the percentage of hemoglobin molecules in the bloodstream that are bound with oxygen. It is classified within the cardiopulmonary physiological domain and is typically measured using a dual-wavelength PPG sensor integrated into a fingertip or earlobe pulse oximeter [58], [59]. While primarily used as a health and safety indicator, SpO_2 can also provide contextual information in learning environments. For instance, a decline in

[4] SDNN reflects overall autonomic variability and is typically calculated from long-term ECG recordings (e.g., 24-hour Holter) for greater accuracy and physiological validity.

[5] RMSSD reflects short-term parasympathetic activity and is suitable for short-duration analyses, such as 1–5 minute assessments.

[6] However, since BP varies more slowly compared to other physiological signals, large deviations beyond normal ranges may warrant additional monitoring to safeguard learner well-being, such as identifying signs of excessive stress.

SpO_2 may suggest poor air circulation or environmental discomfort, prompting adjustments to ventilation or alerting instructors to potential health-related issues affecting learner performance.

- **HR** [beats per minute]: HR is an indicator of autonomic and cardiovascular function, obtained from ECG or PPG sensors worn on the chest or wrist. The HR accelerates with physical exertion or psychological arousal [60]. Psychologically, an elevated HR during a task often reflects increased arousal, stress, or anxiety, mediated by sympathetic nervous system activity [61]. For instance, when a learner is anxious or under mental pressure, the body releases adrenaline, which can raise the pulse even without physical movement.
- **Respiration rate (RR)** [breaths per minute]: RR refers to the number of breaths taken per minute and is categorized within the autonomic/respiratory physiological domain. It can be measured using motion sensors on the chest or abdomen, respiratory belts, or derived indirectly from PPG signal fluctuations [62]. An elevated RR during learning may indicate increased stress or cognitive load [63]. For instance, a student who begins breathing rapidly during a difficult quiz may be experiencing anxiety. Conversely, a slow and steady RR is typically associated with a relaxed physiological state, suggesting low stress or an appropriate level of task demand.
- **Skin temperature** [°C]: Skin temperature refers to the temperature at the surface of the skin and is classified within the autonomic/peripheral physiological domain. It is typically measured using either a non-contact infrared (IR) thermometer or a skin-contact thermistor [64]. Skin temperature provides a subtle physiological marker for detecting stress or discomfort [65], [66]. For example, a sudden drop in finger skin temperature during a cognitively demanding task may indicate rising stress or anxiety, prompting the system to reduce task difficulty or offer supportive interventions.

Table 1 summarizes the characteristics of the physiological signals discussed above, including their typical ranges, PSI statistics, and the types of sensing devices, with an emphasis on their feasibility as wearable solutions for PAL. Specifically, the PSI statistics column provides empirical distributions from the dataset, which can be directly contrasted with the typical ranges: while most signals (e.g., HR, RR, SpO_2, BP, Skin Temp.) fall within the expected physiological intervals, some (e.g., HRV, GSR, and RR) exhibit narrower or shifted distributions, reflecting the limitations of the PSI dataset; the identical ranges reported for EEG bands remain unexplained. In addition, the dataset primarily consists of single-timepoint recordings rather than longitudinal trajectories, which further constrains its ability to capture session-wise dynamics.

TABLE 2. Relevance of physiological signals to learner states in PAL.

Physiological Signal	Cognitive Load	Fatigue	Emotional Arousal
HRV	High [50]	Moderate [68]	High [50]
GSR	Low [13]	–	High [52]
EEG bands	High [54], [55]	High [56]	Moderate [69], [70]
BP	Low [72]	Low [73]	High [74]
SpO_2	Low [75]	Moderate [76]	–
HR	Moderate [23]	Low [71]	High [60]
RR	Moderate [63]	Low [77]	High [78]
Skin Temp.	–	Moderate [79]	High [64]–[66]

High: strong empirical/clinical evidence; **Moderate**: conditionally relevant or task-specific; **Low**: indirect or weak evidence; **--**: no established relationship.

B. RELEVANCE FOR PAL

Based on the physiological signal characteristics described in the previous section, learner states in PAL systems can be broadly categorized into three core dimensions:

- **Cognitive load:** The mental effort required to process and retain instructional content. When excessive, it can overwhelm working memory and degrade performance [19]. Physiological markers include elevated HR, reduced HRV, increased RR, and EEG bands such as suppressed alpha or heightened beta power [21], [22], [67].
- **Fatigue:** A progressive reduction in attentional capacity and mental energy, often leading to disengagement or task avoidance [23], [27]. Indicative physiological signals include increased delta and alpha EEG activity, reduced HRV, lower skin temperature, and diminished GSR amplitude.
- **Emotional arousal:** The physiological activation of the autonomic nervous system in response to stress, anxiety, or relaxation. Arousal related to stress is typically characterized by elevated HR, decreased HRV, increased GSR, accelerated RR, and lowered skin temperature [52], [53], [66], [78]. In contrast, relaxed states are associated with parasympathetic dominance and greater physiological stability.

Table 2 summarizes the empirical and clinical evidence linking specific physiological signals to each of these learner state categories. The referenced studies provide foundational support for applying these signal–state associations in PAL systems. Such physiological indicators enable evidence-informed, real-time difficulty adaptation policies that help maintain learners within optimal arousal zones, mitigate cognitive overload, and support timely instructional interventions.

III. SYSTEM MODEL

We consider a scenario involving N users (or learner groups), indexed by $i \in \{1, \ldots, N\}$. Each user engages in T learning sessions (or instructional tasks), labeled $t = 1, \ldots, T$. In our framework, each session represents a realistic classroom scenario—such as a lecture, group discussion, or exam—where a PAL system adjusts the instructional difficulty in response to the learner's physiological signals. The goal is to optimize individual learning outcomes by accounting for three key learner states—cognitive load, fatigue, and emotional arousal—as discussed in Section II-B.

At the end of each session t, user i is associated with the following variables:

- A performance score $p_t^{(i)}$,
- A level of accumulated fatigue $f_t^{(i)}$,
- A vector of K physiological input modalities, denoted by $\mathbf{s}_t^{(i)} = \left(s_t^{(i,1)}, s_t^{(i,2)}, \ldots, s_t^{(i,K)}\right)$,
- A selected instructional difficulty level $d_t^{(i)}$.

The learner's state $\left(p_{t+1}^{(i)}, f_{t+1}^{(i)}, \mathbf{s}_{t+1}^{(i)}\right)$ for the next session is updated according to a model that accounts for physiological deviation and fatigue accumulation, as described in the following sections. Meanwhile, the instructional difficulty level $d_t^{(i)}$ is selected based on a policy, which may be either rule-based (e.g., threshold-based heuristics) or learned from data using machine learning models.

A. MODELING PERFORMANCE UPDATE IN PAL

In this section, we model how a user's performance $p_t^{(i)}$ evolves over sessions as a function of instructional difficulty $d_t^{(i)}$, physiological state $\mathbf{s}_t^{(i)}$, and fatigue level $f_t^{(i)}$. The performance at session $t+1$ is updated as:

$$p_{t+1}^{(i)} = \mathrm{clip}_{0,100}\Big(p_t^{(i)} + \big[l_{\max} - \big|d_t^{(i)} - l_{t,\mathrm{nor}}^{(i)}\big|\big] - \alpha^{(i)}\,\delta_t^{(i)} - \beta^{(i)} f_t^{(i)} + \gamma^{(i)}\, l_{t,\mathrm{nor}}^{(i)} + n_t^{(i)}\Big). \quad (1)$$

where:

- $l_{\max}$: the maximum difficulty level,
- $l_{t,\mathrm{nor}}^{(i)}$: the reference (or "ideal") difficulty level, which can be adjusted over time as a user-specific optimization parameter based on performance and physiological state,
- $\delta_t^{(i)}$: a distance metric indicating physiological deviation,[7] defined as:

$$\delta_t^{(i)} = \left\| \mathbf{w} \odot \left(\hat{\mathbf{s}}_t^{(i)} - \hat{\mathbf{s}}_{\mathrm{nor}}^{(i)}\right) \right\|_1, \quad (2)$$

  where $\hat{\mathbf{s}}_t^{(i)}$ and $\hat{\mathbf{s}}_{\mathrm{nor}}^{(i)}$ denote the physiological vectors normalized to the [0, 1] range.[8] Here, $\hat{\mathbf{s}}_{\mathrm{nor}}^{(i)}$ represents the baseline vector of physiological signals under normal conditions. This is used to compute deviation from the observed state $\hat{\mathbf{s}}_t^{(i)}$. The weight vector $\mathbf{w} = (w_1, \ldots, w_K)$ reflects the relative importance of each signal modality.[9]
- $\alpha^{(i)}$: user-specific sensitivity to physiological deviation,
- $\beta^{(i)}$: user-specific sensitivity to fatigue,
- $\gamma^{(i)}$: user-specific sensitivity controlling the contribution of the long-term difficulty growth bonus.[10]
- $n_t^{(i)} \sim \mathcal{N}(0, \sigma_n^2)$: zero-mean Gaussian noise modeling residual performance variability (e.g., mood, alertness, or sensor noise).

Over sessions $t = 1, \ldots, T$, the system tracks each user's state $\left(p_t^{(i)}, f_t^{(i)}, \mathbf{s}_t^{(i)}\right)$ and updates performance for PAL using Eq. (1). The term $\alpha^{(i)}\delta_t^{(i)}$ in Eq. (1) models the performance degradation due to physiological deviations, capturing emotional arousal or stress responses. Meanwhile, $\beta^{(i)} f_t^{(i)}$ quantifies the impact of fatigue accumulation. Together, these terms enable the model to capture transient fluctuations in both emotional arousal and cognitive load conditions, which vary over time and across learners. When parameters such as $\alpha^{(i)}$, $\beta^{(i)}$, $\gamma^{(i)}$ and $\mathbf{s}_{\mathrm{nor}}^{(i)}$ are shared across users, the model operates in a shared-parameter mode. Alternatively, in a personalized or group-based setting, these parameters may vary by user or cohort to reflect inter-individual differences.

1) INTERPRETATION

In Eq. (1), the term $l_{\max} - \left|d_t^{(i)} - l_{t,\mathrm{nor}}^{(i)}\right|$ represents a baseline performance gain that is maximized when the selected difficulty is close to the reference level $l_{t,\mathrm{nor}}^{(i)}$. However, this potential gain can be diminished by two major penalty terms:

- Fatigue level ($f_t^{(i)}$): As fatigue accumulates across sessions, its impact increases proportionally, reducing net performance improvement.
- Physiological deviation ($\delta_t^{(i)}$): If the learner's physiological state deviates significantly from baseline, performance degrades accordingly.

In contrast, the bonus term $\gamma^{(i)} l_{t,\mathrm{nor}}^{(i)}$ provides a small positive adjustment that encourages gradual upward shifts in the reference difficulty. This ensures that, beyond reacting to immediate fatigue and physiological states, the system can promote long-term learning progress by rewarding higher baseline difficulty levels. While assigning a fixed difficulty near $l_{t,\mathrm{nor}}^{(i)}$ may yield reasonable performance on average, it neglects the learner's time-varying internal state. Since physiological and cognitive conditions fluctuate over time and across individuals, dynamically adapting the difficulty level enables the system to mitigate temporary disadvantages and better support learning. Therefore, static strategies may be suboptimal; adaptive policies—rule-based or AI-driven—are essential for maximizing performance under diverse and evolving conditions.

B. MODELING FATIGUE AND PHYSIOLOGICAL SIGNAL DYNAMICS IN PAL

1) FATIGUE UPDATE

To model the cumulative and partially recoverable nature of fatigue, we define the following update rule:

$$f_{t+1}^{(i)} = \mathrm{clip}_{0,\,F_{\max}^{(i)}}\left(f_t^{(i)} + c_1^{(i)}\, d_t^{(i)} - c_2^{(i)} - \Delta_{\mathrm{recover}}^{(i)}\left(\mathbf{s}_t^{(i)}\right)\right), \quad (3)$$

[7] Depending on the analysis setting, the deviation term can be computed either from the instantaneous deviation or from a smoothed or aggregated form across sessions. Details of the specific deviation-aggregation configurations used in the sensitivity analysis are provided in Section V.

[8] The physiological signals $\mathbf{s}_t^{(i)} = \left(s_t^{(i,1)}, s_t^{(i,2)}, \ldots, s_t^{(i,K)}\right)$ can be normalized using Min–Max scaling, Z-score standardization, or other methods depending on the data distribution.

[9] $\sum_k w_k = 1$ and $\mathbf{w}$ can be set based on heuristics, feature importance from AI models, or optimized via Bayesian search.

[10] This term encourages upward adjustment of the reference difficulty level to support long-term learning. A small value (e.g., 0.05–0.2) is used to avoid disrupting short-term adaptation.

where $c_1^{(i)}$ and $c_2^{(i)}$ denote user-specific coefficients for fatigue accumulation and recovery, respectively, and $\Delta_{\text{recover}}^{(i)}(\cdot)$ provides an additional recovery adjustment based on physiological signals.[11] $F_{\max}^{(i)}$ is the maximum allowable fatigue level.

a: INTERPRETATION

Eq. (3) balances challenging learning conditions and physiological recovery. Higher difficulty increases fatigue, while recovery consists of a constant term $c_2^{(i)}$ and an adaptive component $\Delta_{\text{recover}}^{(i)}$ modulated by the user's physiological state. This allows modeling user archetypes with varying susceptibility to fatigue and burnout. When selecting $d_t^{(i)}$, the system must consider future fatigue penalties that may hinder performance (see Section III-A). Adaptive policies therefore account for the trade-off between immediate learning gains and longer-term fatigue accumulation.

2) PHYSIOLOGICAL SIGNAL DYNAMICS

To account for temporal changes in the user's physiological signals $\mathbf{s}_t^{(i)} = (s_t^{(i,1)}, s_t^{(i,2)}, \ldots, s_t^{(i,K)})$, we model each component $s_t^{(i,k)}$ using a simple first-order autoregressive process:

$$s_{t+1}^{(i,k)} = s_t^{(i,k)} + \rho_k^{(i)}\left(s_{\text{nor}}^{(i,k)} - s_t^{(i,k)}\right) + \eta_k^{(i)} \cdot \phi\left(d_t^{(i)} - l_{t,\text{nor}}^{(i)}\right) + \epsilon_t^{(i,k)}, \tag{4}$$

where $\rho_k^{(i)}$ is the user-specific return rate toward the baseline value $s_{\text{nor}}^{(i,k)}$; $\eta_k^{(i)}$ denotes the sensitivity of signal k to difficulty-induced arousal or stress, and scales the effect of the deviation from the reference difficulty level through a nonlinear function $\phi(\cdot)$[12]; $\epsilon_t^{(i,k)} \sim \mathcal{N}(0, \sigma_\epsilon^2)$ represents random variability due to physiological fluctuation or sensor noise.

a: INTERPRETATION

Eq. (4) captures the tendency for physiological states to return to their personal baselines unless perturbed by external or task-related stress. It enables the system to maintain continuity of learner state across sessions, providing a smooth latent trajectory of physiological adaptation. This is particularly advantageous given that the PSI dataset [40] primarily consists of single-timepoint recordings rather than longitudinal trajectories; the proposed dynamics model compensates for this limitation by generating session-wise trajectories that approximate realistic temporal evolution of physiological signals.

Together, Eqs. (1), (3), and (4) jointly model the learner's cognitive load (via task difficulty and performance), fatigue (via cumulative effort), and emotional arousal (via physiological deviations), forming an integrated foundation for state-aware adaptation in PAL.

IV. DIFFICULTY ADAPTATION POLICIES FOR PAL

Within a PAL system, the performance update model in Eq. (1) is combined with a policy that selects the instructional difficulty level $d_t^{(i)}$. The difficulty may be defined over either a discrete or continuous range; in this work, we assume a discrete set $d_t^{(i)} \in \{1, \ldots, 5\}$. At the beginning of each session t, the system selects a difficulty level for user i based on the current policy. Various difficulty adaptation policies can be applied under this framework.

A. YERKES–DODSON-BASED POLICY (YDBP)

The Yerkes–Dodson law [41] states that performance is maximized at moderate levels of arousal, whereas both under- and over-arousal impair cognitive functioning. We generalize this principle to multiple physiological input modalities $\mathbf{s}_t^{(i)}$. For each signal $s_t^{(i,k)}$, we define an optimal arousal interval $[L_k, U_k]$ in the original signal space.[13] To account for signal-specific directionality, we introduce the direction-adjusted arousal value as:

$$\tilde{s}_t^{(i,k)} = \alpha_k \cdot s_t^{(i,k)}, \tag{5}$$

where $\alpha_k \in \{-1, +1\}$ is the polarity coefficient for signal k.[14] Accordingly, we define the polarity-adjusted optimal arousal interval in the transformed $\tilde{s}$ domain as $[\tilde{L}_k, \tilde{U}_k] := [\alpha_k \cdot L_k, \alpha_k \cdot U_k]$. To preserve ordering, we assume $\tilde{L}_k < \tilde{U}_k$ by swapping bounds if necessary. Using the adjusted value $\tilde{s}_t^{(i,k)}$, we define the directional deviation $g_t^{(i,k)}$ from the optimal arousal range as:

$$g_t^{(i,k)} = \begin{cases} -\dfrac{\tilde{L}_k - \tilde{s}_t^{(i,k)}}{\tilde{L}_k - \tilde{L}_k^{\min}}, & \text{if } \tilde{s}_t^{(i,k)} < \tilde{L}_k \quad \text{(under-arousal)} \\ 0, & \text{if } \tilde{L}_k \le \tilde{s}_t^{(i,k)} \le \tilde{U}_k \quad \text{(optimal)} \\ \dfrac{\tilde{s}_t^{(i,k)} - \tilde{U}_k}{\tilde{U}_k^{\max} - \tilde{U}_k}, & \text{if } \tilde{s}_t^{(i,k)} > \tilde{U}_k \quad \text{(over-arousal)} \end{cases} \tag{6}$$

Here, $\tilde{L}_k^{\min} := \alpha_k \cdot L_k^{\min}$ and $\tilde{U}_k^{\max} := \alpha_k \cdot U_k^{\max}$ represent the polarity-adjusted normalization bounds. These adjustments normalize $g_t^{(i,k)}$ within $[-1, 1]$, where 0 indicates optimal arousal, negative values indicate under-arousal, and positive values indicate over-arousal.

Let $\mathbf{g}_t^{(i)} = \left(g_t^{(i,1)}, \ldots, g_t^{(i,K)}\right)$ be the vector of directional arousal deviations, and let $\tilde{\mathbf{w}} = (\tilde{w}_1, \ldots, \tilde{w}_K)$ be the corresponding normalized sensitivity weights, satisfying

[11] While $c_2^{(i)}$ ensures a fixed baseline recovery after each session, $\Delta_{\text{recover}}^{(i)}(\mathbf{s}_t^{(i)})$ allows additional recovery when the user's physiological state indicates low stress or stability. This enables more personalized and condition-dependent fatigue modeling.

[12] The function $\phi(\cdot)$ maps difficulty deviation to a bounded excitation value; for example, one may use the sigmoid or hyperbolic tangent function.

[13] The original signal space refers to the untransformed domain of each signal—i.e., the raw measurement scale such as ms for HRV. The typical ranges listed in Table 1 provide reference intervals for interpreting $[L_k, U_k]$ in practice.

[14] For example, $\alpha_k = +1$ for signals such as HR, RR, and GSR, where higher values typically indicate higher arousal. Conversely, for HRV, where higher values imply lower arousal, we set $\alpha_k = -1$.

$\sum_k \tilde{w}_k = 1$.[15] We then define two scalar metrics to quantify multimodal arousal as:

1) TOTAL AROUSAL DEVIATION

$$A_t^{(i)} = \left\| \tilde{\mathbf{w}} \odot \mathbf{g}_t^{(i)} \right\|_1 . \tag{7}$$

This quantity represents the aggregate deviation of user i's physiological state from the optimal arousal range at session t, weighted across all signal modalities. Larger values of $A_t^{(i)}$ indicate a greater overall deviation—regardless of whether it stems from under- or over-arousal.

2) NET DIRECTIONAL AROUSAL

$$Z_t^{(i)} = \frac{\tilde{\mathbf{w}}^\top \mathbf{g}_t^{(i)}}{A_t^{(i)} + \varepsilon}, \tag{8}$$

where $\varepsilon \ll 1$ is a small constant introduced to avoid division by zero. The sign of $Z_t^{(i)}$ indicates the dominant direction of arousal deviation—negative for under-arousal and positive for over-arousal. Its magnitude reflects the degree of directional imbalance: values close to zero indicate a balanced deviation across signals, while values near ± 1 suggest a strongly unidirectional deviation.

Using the two metrics defined in Eqs. (7) and (8), the difficulty level $d_t^{(i)}$ is determined by the following piecewise rule, referred to as the Yerkes–Dodson-Based Policy (YDBP):

$$d_t^{(i)} = \begin{cases} 2, & \text{if } A_t^{(i)} < \tau_{\text{low}} \text{ or } A_t^{(i)} > \tau_{\text{high}}, \\ 3, & \tau_{\text{low}} \le A_t^{(i)} \le \tau_{\text{high}} \text{ and } |Z_t^{(i)}| > \zeta, \\ 4, & \tau_{\text{low}} \le A_t^{(i)} \le \tau_{\text{high}} \text{ and } |Z_t^{(i)}| \le \zeta. \end{cases} \tag{9}$$

a: INTERPRETATION

- Extremely low or high arousal ($A_t^{(i)} < \tau_{\text{low}}$ or $> \tau_{\text{high}}$): The user's physiological state indicates either under-stimulation or excessive stress. To mitigate disengagement or overload, the system selects a low difficulty level ($d_t^{(i)} = 2$).
- Unbalanced moderate arousal ($A_t^{(i)}$ within range, but $|Z_t^{(i)}| > \zeta$): Although the overall deviation is moderate, the signal pattern is skewed toward under- or over-arousal. A mid-level difficulty ($d_t^{(i)} = 3$) is chosen to promote rebalancing.
- Balanced optimal arousal ($A_t^{(i)}$ within range and $|Z_t^{(i)}| \le \zeta$): The user is physiologically stable and within the optimal arousal zone. A higher difficulty level ($d_t^{(i)} = 4$) is provided to encourage cognitive growth.

Eq. (9) formalizes the Yerkes–Dodson law [41] as a three-level, discrete adaptation policy. The underlying principle of YDBP is that learning performance peaks within a moderate arousal zone. By adjusting task difficulty—raising it when arousal is low and lowering it when arousal is excessive—the system aims to maintain optimal engagement and cognitive efficiency. While this discretized implementation simplifies the original inverted-U relationship,[16] prior studies have demonstrated its effectiveness in aligning instructional demand with learners' physiological states [80], [81], [82], [83].

B. DISTANCE-BASED POLICY (DBP)

The distance-based approach is a simple rule-based policy that adapts learning difficulty according to how far the user's current physiological state deviates from a defined baseline. The underlying assumption is that users who are closer to their baseline state are better equipped to handle more cognitively demanding tasks, while those exhibiting greater deviations may benefit from reduced difficulty to prevent excessive cognitive or physiological strain.

Based on the deviation metric in Eq. (2), the Distance-Based Policy (DBP) determines the difficulty level $d_t^{(i)}$ as a piecewise function using two thresholds, C_1 and C_2, to categorize low, moderate, and high deviation levels:

$$d_t^{(i)} = \begin{cases} 4, & \text{if } \delta_t^{(i)} < C_1, \\ 3, & \text{if } C_1 \le \delta_t^{(i)} < C_2, \\ 2, & \text{otherwise.} \end{cases} \tag{10}$$

1) INTERPRETATION

- Small deviation ($\delta_t^{(i)} < C_1$): The user's physiological state is close to their normal baseline. Accordingly, the system assigns a high difficulty level ($d_t^{(i)} = 4$), assuming readiness for increased cognitive demand.
- Moderate deviation ($C_1 \le \delta_t^{(i)} < C_2$): The deviation is noticeable but not extreme. A medium difficulty level ($d_t^{(i)} = 3$) is chosen to support steady engagement without causing strain.
- Large deviation ($\delta_t^{(i)} \ge C_2$): Significant deviation may indicate elevated stress, fatigue, or other atypical states. The system lowers the difficulty ($d_t^{(i)} = 2$) to reduce the risk of overload.

From a pedagogical standpoint, DBP aims to dynamically align task difficulty with the learner's physiological readiness. It seeks to avoid both under-challenge (when the user is near baseline) and over-challenge (when signs of strain are present), thereby supporting personalized pacing. Compared to the YDBP (see Section IV-A), which relies on categorical thresholds to detect under- or over-arousal states, the distance-based approach treats deviation as a continuous scalar. This enables smoother and more fine-grained adaptation, potentially offering better responsiveness to gradual changes in the learner's condition.

C. AI-BASED POLICY (AIBP)

The AI-Based Policy (AIBP) is a data-driven approach that selects instructional difficulty levels by predicting the cumulative impact of future performance and fatigue over

[15] Weights reflect the relative relevance of each signal to arousal, as summarized in Table 2.

[16] The inverted-U relationship describes how performance improves with increasing arousal up to a moderate point, after which further arousal impairs performance.

a finite planning horizon. For each user i at session t, the policy evaluates all candidate difficulty sequences $(d_t, d_{t+1}, \ldots, d_{t+H-1}) \in \mathcal{D}^H$ over a prediction horizon H, where $\mathcal{D} = \{1, \ldots, 5\}$. Starting from the current state $(p_t^{(i)}, \mathbf{s}_t^{(i)}, f_t^{(i)})$, the AIBP uses learned transition models to simulate the resulting sequence of physiological states, fatigue, and performance values based on Eqs. (1)–(4).

The optimal sequence is chosen to minimize a cumulative cost function that balances performance improvement against fatigue accumulation:

$$(d_t^{(i)}, \ldots, d_{t+H-1}^{(i)})^* = \arg \min_{(d_t,\ldots,d_{t+H-1})\in\mathcal{D}^H} \sum_{h=0}^{H-1} \Big[-p_{t+h+1}^{(i)} + \lambda f_{t+h+1}^{(i)} \Big], \quad (11)$$

where $\lambda > 0$ is a user-specific trade-off parameter. In practice, only the first difficulty level $d_t^{(i)}$ of the optimal sequence is applied:

$$d_t^{(i)} = \text{first element of } (d_t, \ldots, d_{t+H-1})^*. \quad (12)$$

To support this look-ahead policy, we train predictive models via supervised learning to simulate one-step transitions of the form:

$$\left(p_t^{(i)},\, \mathbf{s}_t^{(i)},\, f_t^{(i)},\, d_t^{(i)}\right) \rightarrow \left(p_{t+1}^{(i)},\, \mathbf{s}_{t+1}^{(i)},\, f_{t+1}^{(i)}\right), \quad (13)$$

where $p_{t+1}^{(i)}$ is computed from Eq. (1), and the other next-state variables are given by Eqs. (4) and (3). Letting ψ denote the parameters of this one-step simulator G_ψ, we minimize the prediction loss over all sessions[17] as:

$$\mathcal{L}_{\text{sim}}^{(i)} = \frac{1}{T} \sum_{t=1}^{T} \left\| G_\psi(p_t^{(i)},\, \mathbf{s}_t^{(i)},\, f_t^{(i)},\, d_t^{(i)}) - (p_{t+1}^{(i)},\, \mathbf{s}_{t+1}^{(i)},\, f_{t+1}^{(i)}) \right\|^2. \quad (14)$$

This formulation enables the AIBP to anticipate future outcomes of difficulty choices, adapt to the learner's evolving physiological and cognitive state, and support horizon-aware personalized adaptation.

We implement two widely-used regression models to instantiate the one-step simulator $G_{\psi^{(i)}}$, which maps the current state and difficulty level to the predicted next-step outcomes, including performance, physiological state, and fatigue. The simulator is trained on synthetic trajectories generated from the domain equations (Eqs. (1)–(4)), making it a domain-informed transition model.

[17] In the first session, the difficulty level $d_1^{(i)}$ is initialized using the self-reported cognitive load labels (Low, Moderate, High) in the PSI dataset, mapped to discrete difficulty levels 2, 3, and 4, respectively. For all subsequent sessions ($t \geq 2$), the difficulty level $d_t^{(i)}$ is determined by the AIBP, enabling dynamic adaptation based on the learner's evolving state.

1) RANDOM FOREST (RF)

A Random Forest regressor [47] ensembles multiple decision trees and outputs their average prediction. We use independent RF models to predict each of the next-step variables $p_{t+1}^{(i)}$, $\mathbf{s}_{t+1}^{(i)}$, and $f_{t+1}^{(i)}$ given the current state $(p_t^{(i)}, \mathbf{s}_t^{(i)}, f_t^{(i)})$ and difficulty level $d_t^{(i)}$. Random Forest is robust to nonlinearities and noise and offers interpretable feature importance, making it well-suited for modeling heterogeneous learner states.

2) XGBoost (XGB)

XGBoost [48] is a gradient boosting framework that sequentially fits decision trees to residual errors. To capture complex nonlinear interactions among performance, physiological signals, fatigue, and instructional difficulty, we train separate XGBoost regressors for each target dimension in the simulator output. XGBoost is particularly effective for tabular data and provides strong generalization performance under limited sample regimes.

For both model types, we apply the learned simulator G_ψ recursively over a planning horizon H during multi-step rollout, enabling the AIBP to anticipate future trajectories resulting from candidate difficulty sequences.

a: INTERPRETATION

A key advantage of the AIBP framework—particularly in its multi-step form—is that it does not require ground-truth measurements of performance or fatigue. Since the PSI dataset [40] does not contain explicit performance scores and fatigue levels, the target variables $(p_{t+1}^{(i)}, \mathbf{s}_{t+1}^{(i)}, f_{t+1}^{(i)})$ are synthesized using the simulation-based update models defined in Eqs. (1)–(4). This design enables training on realistic physiological input while preserving theoretical consistency with the PAL framework. The resulting learning process constitutes a form of simulation-driven supervised learning, in which structured pseudo labels are generated from a domain-informed transition model $G_{\psi^{(i)}}$. By learning from these synthetic trajectories, the AIBP generalizes how learner performance and internal state evolve under different physiological conditions and instructional difficulty levels. During inference, the policy evaluates candidate difficulty sequences via horizon-based simulation, supporting personalized and forward-looking adaptation.

Despite its flexibility, the effectiveness of this approach depends on several key factors:

- Fidelity of the update models: If Eqs. (1)–(4) poorly approximate actual learner dynamics, the pseudo labels may be biased and mislead the model.
- Training data diversity: Limited coverage in $(p_t^{(i)}, \mathbf{s}_t^{(i)}, f_t^{(i)}, d_t^{(i)})$ combinations may restrict generalization and increase overfitting risk.
- Robustness to unseen states: When a user's physiological profile deviates significantly from the training distribution, the policy may struggle with extrapolation or produce unstable predictions.

Unlike rule-based policies such as YDBP and DBP, which rely on predefined heuristics, the AIBP learns from data to infer the nuanced interplay between difficulty, physiology, and learning outcomes. While this data-driven design enables finer personalization and temporal adaptation, it also necessitates adequate training coverage and computational capacity to reliably estimate the simulator $G_{\psi^{(i)}}$ and optimize the resulting policy.

V. EXPERIMENTS

In this section, we evaluate and analyze the performance of the YDBP, DBP, and AIBP policies introduced in Section IV, using the physiological signals from the PSI dataset [40] described in Section II, based on the system model presented in Section III.

A. EXPERIMENTAL SETTING

1) DATASET

Although the PSI dataset provides multiple physiological signals as depicted in Table 1, in our experiments we focus only on a subset $\mathbf{s}_t^{(i)} = \left(s_t^{(i,1)}, s_t^{(i,2)}, s_t^{(i,3)}\right)$, corresponding to HRV, HR, and RR, respectively. This selection is motivated by their practicality, as these signals can be reliably obtained from commercially available wearable devices such as smartwatches, making our experimental setup more aligned with realistic deployment scenarios of PAL systems.

To ensure that all difficulty adaptation policies in Section IV start from the same initial state, we constructed the initial dataset for each learner as $\left(p_0^{(i)}, f_0^{(i)}, \mathbf{s}_0^{(i)}\right)$. Since no ground-truth data were available for the initial performance and fatigue level, we set $p_0^{(i)} = 50$ (within the 0–100 range) and $f_0^{(i)} = 5$ (within the 0–10 range) as fixed initial values, corresponding to neutral mid-range baselines in their respective scales. The initial physiological state vector $\mathbf{s}_0^{(i)}$ was configured using the values of HRV, HR, and RR from the PSI dataset corresponding to 1,000 learners. Furthermore, the initial difficulty level $d_0^{(i)}$ was determined by mapping the self-reported cognitive load labels in the PSI dataset (Low, Moderate, High) to difficulty levels 2, 3, and 4, respectively.

Subsequently, each learner's trajectory $\left(p_{t+1}^{(i)}, f_{t+1}^{(i)}, \mathbf{s}_{t+1}^{(i)}\right)$ is generated and updated according to Eqs. (1), (3), and (4), with the difficulty level $d_{t+1}^{(i)}$ determined by the updated learner state and the corresponding adaptation policy. By iterating this process, we synthesize session-wise trajectories that expand the PSI dataset into 50,000 learner–session–difficulty observations (1,000 learners $\times$10 sessions $\times$5 difficulty levels). This procedure can be regarded as model-based synthetic trajectory generation, as new session-level data points are produced beyond the original PSI dataset by applying the update equations. No missing values or outliers were present in the PSI dataset, and the signals are normalized as described in Eq. (2).

Instead of directly using Eq. (1) to generate pseudo labels, we adopt a Kalman filtering–based labeling mechanism [84], [85], [86] to mitigate potential circularity in the AIBP. Unlike the deterministic update equation, the Kalman-based label is obtained from the *posterior* update of a Kalman filter, which incorporates latent-state estimation, stochastic innovation, and observation uncertainty. This produces pseudo labels that are partially decoupled from the closed-form simulation model, providing a more robust and less model-dependent supervisory signal for AIBP training.

For the evaluation of AIBP, we adopt group k-fold cross-validation by learner with $k = 20$. This choice was made to reduce the variance of the validation folds, while maintaining approximately 50 learners per fold. In each fold, 50 learners (approximately 2,500 observations) are held out for validation, while the remaining 950 learners are used for training, ensuring that no learner's history appears in both training and validation (leakage prevention). The resulting 50,000 observations and cross-validation protocol are specifically employed for assessing the performance of the proposed AIBP.

2) PARAMETER ASSUMPTIONS FOR THE SYSTEM MODEL AND POLICIES

In this subsection, we specify the parameter assumptions used in the system model equations and in the adaptive difficulty adjustment policies. These assumptions define initial conditions, scaling factors, and control variables, and are essential for reproducing the system dynamics and ensuring a fair comparison across policies.

- **Parameters in Eq. (1)**: Except for the sensitivity analysis[18] presented in Fig. 2 and Table 4, the deviation weighting factor, fatigue penalty sensitivity, and difficulty-alignment reward coefficient were fixed at $\alpha^{(i)} = 1$, $\beta^{(i)} = 0.2$, and $\gamma^{(i)} = 0$, respectively. Setting $\alpha^{(i)} = 1$ ensured that the physiological deviation term $\delta_t^{(i)}$ was fully reflected as a performance penalty. The choice $\beta^{(i)} = 0.2$ kept the fatigue penalty comparable in scale to the deviation penalty $(\alpha^{(i)}\delta_t^{(i)})$, preventing excessive suppression of short-term learning progress while still capturing the cumulative effects of fatigue over sessions. The optimal difficulty level was fixed at $l_{t,\text{nor}}^{(i)} = 3$ across all sessions. Although $\gamma^{(i)} = 0$ has no effect in this setting, the term is retained for interpretability and can be extended to $\gamma^{(i)} > 0$ when personalized or time-varying $l_{t,\text{nor}}^{(i)}$ is considered.
- **Weights in Eq. (2)**: For YDBP and DBP, uniform weights were applied across modalities. For AIBP, Shapley Additive Explanations (SHAP) [109] were applied to the learned transition simulator G_ψ to quantify the contribution of each physiological signal, and the resulting importance scores were directly used as weights ($w_{\text{HRV}} = 0.1608$, $w_{\text{HR}} = 0.4456$, $w_{\text{RR}} = 0.3937$).

[18] Fig. 2 and Table 4 present a sensitivity and ablation analysis in which α is systematically varied across a wide range of values, in contrast to the fixed baseline configuration.

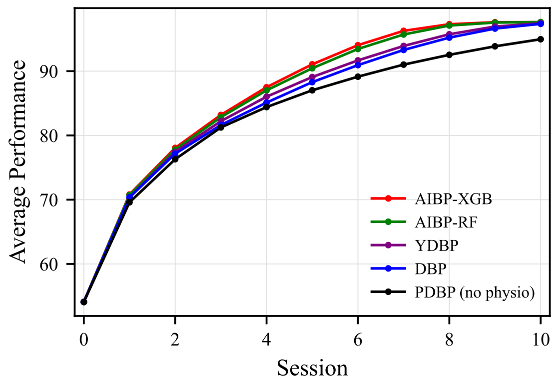

FIGURE 1. **Session-wise average performance across adaptive difficulty policies.**

- **Parameters in Eq. (3):** To reflect the asymmetry between accumulation and recovery at the session level, we set $c_1^{(i)} = 0.5$ and $c_2^{(i)} = 0.2$. The additional recovery term $\Delta_{\text{recover}}^{(i)}$ was activated when HR, HRV, and RR were within their normal ranges, allowing physiological stability to reduce fatigue. The maximum fatigue level was fixed at $F_{\max}^{(i)} = 10$.
- **Parameters in Eq. (4):** The recovery rate $\rho_k^{(i)}$ and the noise scaling factor $\epsilon_k^{(i)}$ were set according to signal-specific response characteristics: HR: $\rho_k^{(i)} = 0.3$, $\epsilon_k^{(i)} \in [0.05, 0.10]$; RR: $\rho_k^{(i)} = 0.2$, $\epsilon_k^{(i)} \in [0.03, 0.05]$; HRV: $\rho_k^{(i)} = 0.1$, $\epsilon_k^{(i)} \in [0.02, 0.05]$. The sensitivity to difficulty changes was set to $\eta_{\text{HRV}} = 0.5$ and $\eta_{\text{HR}} = \eta_{\text{RR}} = 0.25$, reflecting the relevance to cognitive load summarized in Table 2. The activation function $\phi(\cdot)$ was implemented as the hyperbolic tangent function (tanh).
- **Parameters in YDBP and DBP:** For YDBP, the optimal arousal bounds in Eq. (9) were set to HRV: [20, 70], HR: [60, 100], and RR: [12, 20], with normal values defined as the midpoint of each range (HRV = 45, HR = 80, RR = 16). The normalization intervals $[L_k^{\min}, U_k^{\max}]$ were derived from the PSI dataset as HRV: [17.41, 78.84], HR: [60, 99], and RR: [12, 24]. Uniform weights were applied in Eq. (7), and the thresholds in Eq. (9) were set to $\tau_{\text{low}} = 0.1$, $\tau_{\text{high}} = 0.2$, and $\zeta = 0.6$. For DBP, the deviation thresholds in Eq. (10) were set to $C_1 = 0.2$ and $C_2 = 0.5$, representing average deviations of 20% and 50% from the normal physiological baseline.
- **Parameters in AIBP:** The planning horizon in Eq. (11) was set to $H = 2$, which balances short-term responsiveness with the ability to anticipate fatigue accumulation, while avoiding excessive computational cost and error propagation in longer rollouts. The trade-off parameter was fixed at $\lambda = 1.0$, in Eq. (11) chosen heuristically to balance performance gain and fatigue penalty on comparable scales. The one-step simulator G_ψ was trained on the synthetic dataset described in Section V-A. The RF model was configured with 50 trees in the ensemble, while XGB was configured with 50 boosting rounds, a maximum depth of 4, and a learning rate of 0.1.

B. POLICY EVALUATION AND ANALYSIS

1) SESSION-WISE PERFORMANCE TRENDS

In this subsection, we analyze how each adaptive difficulty adjustment policy influences learner performance as sessions progress. Session-wise performance trajectories serve as key indicators for evaluating long-term learning efficiency, adaptation speed, and final achievement level. Early-session improvement patterns reflect the initial suitability of difficulty adjustment strategies and learner engagement, whereas late-session performance levels indicate the cumulative learning effects provided by each policy.

Fig. 1 illustrates the session-wise average performance trajectories across the adaptive policies, and Table 3 summarizes the corresponding Area Under the receiver operating characteristic Curve (AUC) metrics. All policies exhibit steady monotonic improvement as the number of sessions increases, confirming that adaptive difficulty adjustment effectively enhances learner performance under the Kalman filtering–based pseudo-labeling regime. The baseline policy, Performance-Deviation-Based Policy (PDBP, no physio), uses neither physiological signals nor fatigue information; instead, it adjusts difficulty solely based on performance-deviation heuristics. This baseline enables clear isolation of the benefit contributed by physiology-informed signals within the adaptive learning framework.

To clarify the planning horizon of each adaptive policy, we note that YDBP and DBP operate as myopic, one-step policies that adjust difficulty solely based on the current physiological and fatigue state without performing any explicit multi-step forecasting. In contrast, the AI-based policies (AIBP-RF and AIBP-XGB) are implemented as lookahead policies with a prediction horizon $H = 2$, as described in Section IV-C: each candidate difficulty level is evaluated by simulating the learner's predicted trajectory over the next two sessions using the learned one-step transition models. PDBP, while not physiology-based, functions as a performance-deviation band controller that adjusts difficulty one step ahead based on short-term fluctuations in observed performance, without involving any multi-step prediction.

Among the adaptive policies, AIBP-XGB and AIBP-RF consistently achieve the strongest performance throughout the learning sessions. As shown in Table 3, both AIBP variants yield the highest AUC values across all three session windows. During the early stage (Sessions 0–3), AIBP-XGB reaches an AUC of 77.53 and AIBP-RF reaches 77.28, corresponding to improvements of +1.67 and +1.42 over the baseline (PDBP), respectively. This advantage becomes even more pronounced in Sessions 4–7, where AIBP-XGB and AIBP-RF achieve the largest gains—92.33 (+4.37) and 91.76 (+3.80). The same pattern persists in Sessions 8–10, with AIBP-XGB reaching 97.54 (+3.73) and AIBP-RF reaching 97.46 (+3.65), again outperforming YDBP and DBP by a clear margin.

TABLE 3. AUC performance across sessions and improvements over the baseline (PDBP, no physio).

Policy	AUC Mean (Sessions 0–3) (value; ΔAUC vs. Baseline)	AUC Mean (Sessions 4–7) (value; ΔAUC vs. Baseline)	AUC Mean (Sessions 8–10) (value; ΔAUC vs. Baseline)	First Session Score ≥ 90
AIBP-XGB	77.53 (+1.67)	92.33 (+4.37)	97.54 (+3.73)	5
AIBP-RF	77.28 (+1.42)	91.76 (+3.80)	97.46 (+3.65)	5
YDBP	76.85 (+0.99)	90.24 (+2.28)	96.77 (+2.96)	6
DBP	76.58 (+0.72)	89.48 (+1.52)	96.46 (+2.65)	6
Baseline	75.86 (0.00)	87.96 (0.00)	93.81 (0.00)	7

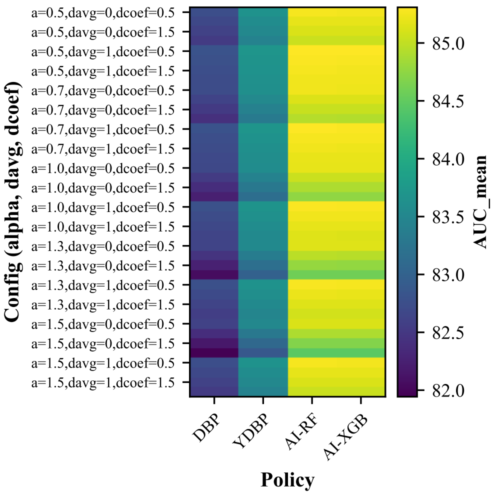

FIGURE 2. Sensitivity Analysis of AUC Performance Across Policy Types With Respect to Configuration Parameters (α, d_{avg}, d_{coef}).

In contrast, the rule-based policies (YDBP and DBP) produce more moderate improvements over the baseline. YDBP attains AUC values of 76.85, 90.24, and 96.77 across the three windows, while DBP yields 76.58, 89.48, and 96.46—each showing smaller gains than the AI-based approaches. Furthermore, the rate of early adaptation differs noticeably across policy types: both AIBP-XGB and AIBP-RF surpass the 90-point performance threshold by Session 5, whereas YDBP and DBP require Session 6, and the PDBP (no physio) reaches the same threshold only by Session 7.

Overall, these results demonstrate that AIBP-based adaptive learning provides (i) superior performance across sessions, (ii) larger AUC improvements relative to the baseline, and (iii) faster early-session convergence—reinforcing the robustness of AI-driven adaptation even under the proposed pseudo-labeling mechanism.

Fig. 2 presents the extended sensitivity and ablation analysis conducted to examine the impact of the coefficient α—the key parameter governing the influence of physiological deviations on performance degradation in Eq. (1).[19] Because $\alpha = \delta_{\text{coef}} \times \delta_t$, its magnitude reflects both the severity of moment-to-moment physiological deviation (δ_t) and the deviation-scaling factor (δ_{coef}). To systematically assess how variations in this parameter influence policy performance, we evaluated a wide configuration space consisting of $\alpha \in \{0.5, 0.7, 1.0, 1.3, 1.5\}$, $\delta_{\text{coef}} \in \{0.5, 1.0, 1.5, 2.0\}$, and $\delta_{\text{avg}} \in \{\text{False}, \text{True}\}$. Here, δ_{avg} is a binary flag used only in the sensitivity analysis to indicate whether physiological deviation is computed solely from the instantaneous deviation δ_t or from a combined deviation measure that incorporates both instantaneous and session-averaged values. This configuration allows us to evaluate the robustness of the performance update model under different deviation-aggregation strategies.

In Fig. 2, each row in the heatmap corresponds to one configuration from this parameter space, and the resulting AUC performance is compared across the four adaptive policy types (DBP, YDBP, AIBP-RF, and AIBP-XGB). The analysis shows that while all policies exhibit sensitivity to changes in α, the AI-based strategies (AIBP-RF and AIBP-XGB) consistently achieve higher AUC values across the entire configuration set. This indicates that AI-based PAL policies are more robust to deviation-scaling effects and maintain superior performance even under substantial variations in physiological deviation parameters.

TABLE 4. Pairwise Comparisons with Bonferroni Correction. All pairwise differences were statistically significant at $p < 0.001$ (Bonferroni adjusted).

Source	Degrees of Freedom	F-statistic	η_G^2
Policy	3, 2997	1174.22	0.463
Parameter Set	39, 38961	7424.49	0.011
Interaction	117, 116883	162.49	0.00004

TABLE 5. Pairwise Comparisons with Bonferroni Correction. All pairwise differences were statistically significant at $p < 0.001$ (Bonferroni adjusted).

Policy A	Policy B	Cohen's d
AIBP-XGB	AIBP-RF	0.002
AIBP-XGB	YDBP	1.667
AIBP-XGB	DBP	2.026
AIBP-RF	DBP	2.009
AIBP-RF	YDBP	1.652
YDBP	DBP	0.362

2) STATISTICAL VALIDATION

To statistically validate the performance differences observed in the sensitivity and ablation analysis, we conducted a repeated-measures ANOVA across the full parameter-configuration space visualized in Fig. 2. The analysis considered three factors: the policy factor, the parameter-set factor derived from $\alpha = \delta_{\text{coef}} \times \delta_t$ and its associated configuration flags, and the interaction between them. The

[19]Although other parameters such as the fatigue coefficient β and the difficulty-alignment coefficient γ in Eq. (1) may also influence performance dynamics, α directly modulates the core physiology-informed mechanism in our model. For this reason, we prioritized the ablation of α in this study.

results, summarized in Table 4, reveal three significant effects (all $p < 0.001$).

First, the main effect of policy is highly significant ($F(3, 2997) = 1174.22$, $\eta_G^2 = 0.463$), indicating that the choice of adaptive strategy explains a substantial proportion of the variance in AUC performance. This confirms that the AI-based policies (AIBP-RF, AIBP-XGB) consistently outperform rule-based strategies across the entire configuration space. Second, the main effect of the parameter set is statistically significant ($F(39, 38961) = 7424.49$, $\eta_G^2 = 0.011$), although the effect size is considerably smaller. This suggests that variations in α, δ_{avg}, and δ_{coef} have a limited impact on performance, and their contribution remains minor relative to the impact of the policy factor. Third, the interaction between policy and parameter set is significant but exhibits a negligible effect size ($F(117, 116{,}883) = 162.49$, $\eta_G^2 = 0.00004$). This indicates that although performance differences vary slightly across parameter configurations, the superiority of the AI-based policies remains stable and robust under diverse modeling assumptions.

To further examine pairwise differences among the adaptive policies, we conducted Bonferroni-corrected post-hoc comparisons, as summarized in Table 5. All pairwise contrasts were statistically significant at $p < 0.001$, indicating that each policy differed meaningfully from the others. The effect sizes (Cohen's d) show clear performance stratification among the policies. The difference between AIBP-XGB and AIBP-RF was negligible ($d = 0.002$), suggesting that the two AI-based approaches perform equivalently. In contrast, AIBP-XGB achieved substantially higher performance than both YDBP ($d = 1.667$) and DBP ($d = 2.026$), reflecting very large effect sizes. AIBP-RF exhibited a similar pattern, outperforming YDBP ($d = 1.652$) and DBP ($d = 2.009$) with very large effects. Finally, YDBP demonstrated a modest but statistically significant advantage over DBP ($d = 0.362$), corresponding to a small-to-moderate effect size. Taken together, these results provide strong statistical evidence that both AI-based policies (AIBP-XGB and AIBP-RF) deliver substantially higher learner performance than the rule-based policies (YDBP and DBP), while also performing similarly to each other.

3) PHYSIOLOGICAL SIGNAL DYNAMICS UNDER PAL

Beyond performance outcomes, it is also essential to examine how adaptive policies influence learners' physiological states, since maintaining physiological stability and reducing fatigue are critical for sustainable engagement in PAL systems. Analyzing the dynamics of HRV, HR, RR, and fatigue provides complementary evidence on whether the observed performance gains are achieved in alignment with healthy physiological responses.

Fig. 3 provides a consolidated overview of how key physiological modalities (HRV, HR, RR, and fatigue) evolve across sessions under different adaptive policies. The visualization highlights policy-specific patterns, enabling direct comparison between rule-based (YDBP, DBP) and AI-based (AIBP-RF, AIBP-XGB) strategies in terms of how quickly and consistently they guide physiological states toward their normal reference ranges. Several important quantitative trends can be observed. First, HRV exhibits a gradual decline across all policies. For instance, under DBP, HRV decreases from approximately 50 at session 1 to below 46 by session 10, whereas AIBP-RF and AIBP-XGB maintain higher levels, converging around 47 at session 10. This suggests that AIBP policies are more effective in preserving autonomic balance, indicating reduced physiological strain during learning. Meanwhile, HR increases from about 78.8 at session 1 to nearly 80.5 by session 10, with AIBP variants showing faster stabilization toward the normal HR range ($\approx$ 80 bpm) compared to YDBP and DBP. Similarly, RR declines from 18 at session 1 to near 16 by session 10, closely aligning with the normal reference ($\approx$ 16 breaths/min). AIBP policies again demonstrate a more consistent convergence to the physiological norm.

Second, fatigue levels consistently decline across all policies but at different rates. AIBP-XGB reduces fatigue from about 5 at session 1 to nearly 0 by session 6, and AIBP-RF follows a similar trajectory, reaching near-zero by session 6. Incontrast, YDBP and DBP show a slower reduction, approaching 0 by session 8. This indicates that AIBP policies are more effective in mitigating learner fatigue during adaptive learning. Therefore, these results suggest that PAL guided by AIBP not only enhances learning performance but also promotes favorable physiological responses, achieving closer alignment with normal signal ranges and faster reduction of fatigue compared to YDBP and DBP.

4) EXPLAINABILITY OF AIBP

To enhance the interpretability of AIBP decisions, we employed SHAP analysis to quantify the contribution of each physiological feature to the model output. Fig. 4 presents the SHAP summary plot for the three physiological signals—HR, RR, and HRV—illustrating their relative contributions to the predicted learning performance within the AIBP framework.

In Fig. 4, HR exhibits the largest SHAP distribution, confirming that it is the most influential modality. Low HR values (blue) cluster strongly in the negative SHAP region, suggesting that insufficient physiological activation is associated with reduced performance. In contrast, moderately elevated HR values (red) appear near zero or in the positive region, indicating that mild sympathetic activation supports short-term learning engagement. This pattern is consistent with empirical evidence that HR shows high sensitivity to moment-to-moment cognitive load fluctuations and often outperforms HRV in short-window load estimation tasks [87].

RR shows a distinct but physiologically coherent pattern. Lower RR values (blue) appear predominantly within the positive SHAP region, whereas higher RR values (red) shift toward negative contributions. This suggests that controlled and stable respiration facilitates better learning performance,

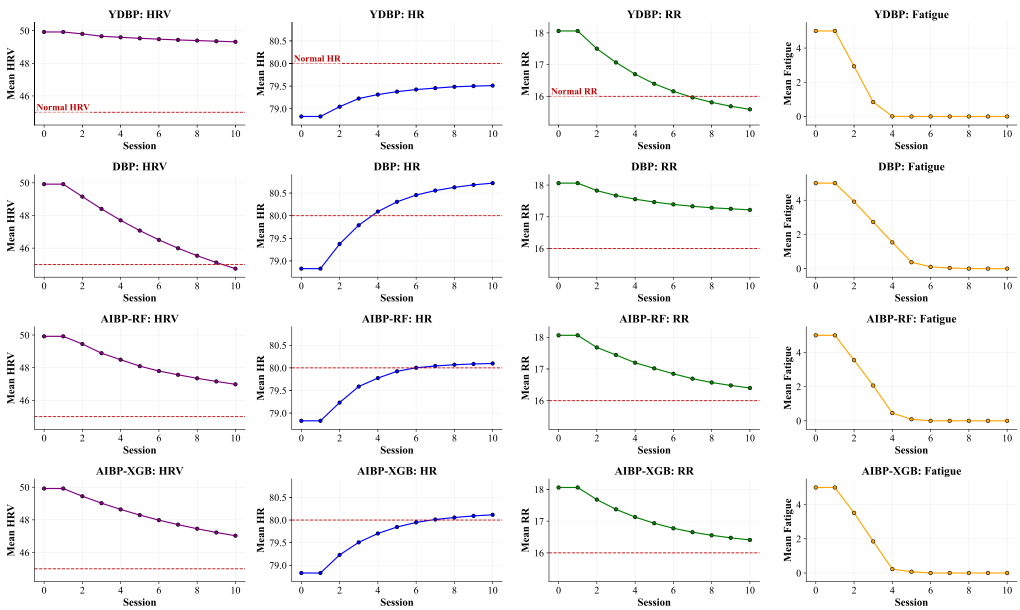

FIGURE 3. Session-wise dynamics of physiological signals (HRV, HR, and RR) and fatigue under different adaptive policies. Red dashed lines indicate normal reference values for each physiological modality.

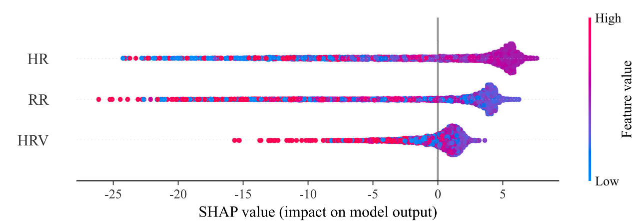

FIGURE 4. SHAP analysis of physiological signal contributions (HR, RR, HRV) to predicted performance in AIBP.

while elevated respiration typically reflects increased stress or cognitive overload that detracts from performance. Prior psychophysiological studies similarly identify RR as a rapid and reliable indicator of short-term workload and attentional dynamics [23].

HRV demonstrates the smallest SHAP magnitude, reflecting both its narrower dynamic range in the dataset and its role as a slower parasympathetic regulatory measure rather than a marker of rapid cognitive fluctuations. Higher HRV values (red) frequently fall within the negative SHAP region, whereas lower HRV values cluster near zero or exhibit mildly positive contributions. This aligns with established interpretations that elevated HRV often corresponds to relaxed or recovery-oriented physiological states that are not indicative of immediate task-focused engagement. Recent educational and psychophysiological research also reports that HRV exhibits limited concurrent validity as a short-window cognitive or attentional state indicator [23], [87], [88].

Overall, the SHAP analysis suggests that performance-enhancing physiological states correspond to (i) moderately elevated HR, (ii) lower RR, and (iii) non-excessively high HRV—together reflecting a physiological profile of moderate arousal and controlled respiration that aligns well with established psychophysiological interpretations of effective short-term cognitive engagement [23], [87], [88].

VI. DISCUSSION

In this section, we discuss the key challenges and future directions for enabling the real-world deployment of AI-driven and physiology-informed PAL systems.

A. PHYSIOLOGICAL SIGNAL RELIABILITY AND ACCESSIBILITY

Wrist-worn devices such as smartwatches equipped with PPG sensors are increasingly used to acquire real-time physiological data in unobtrusive, everyday settings. Recent validation studies have demonstrated that commercial devices such as the Apple Watch, Fitbit, and Garmin can achieve Mean Absolute Percentage Errors (MAPE) below 3% for HR estimation at rest or during low-intensity activities [89], [90]. For HRV, which is more sensitive to motion artifacts and signal noise, PPG-derived indices such as RMSSD and SDNN show strong correlations with gold-standard ECG-based measurements in stationary conditions (Pearson $r > 0.9$) [91], [92]. However, accuracy tends to degrade under movement or high-stress scenarios. In physiology-informed PAL applications—where reliable inference of moment-to-moment fluctuations in cognitive load and fatigue is critical—these wearable devices offer a practical foundation for real-world deployment. Nonetheless, several limitations remain. Motion-induced noise, poor sensor-skin contact, and device variability can impair the robustness of physiological feedback during active learning sessions [93], [94]. In addition, HRV metrics derived from PPG may diverge from ECG-based measurements, particularly for short-time window analyses or during transient physiological events [95]. To mitigate these issues, PAL systems utilizing smartwatch-compatible physiological signals should incorporate signal quality assessment modules, motion artifact reduction techniques, and adaptive filtering strategies to ensure reliability in dynamic environments. Moreover, as emphasized in recent AI-driven biomedical signal studies [42], [44], effectively handling noise and artifacts in real physiological measurements remains a critical direction for future empirical work, particularly when transitioning from simulation-based validation to deployments involving real-world sensing conditions.

Beyond technical considerations, the deployment of physiology-informed PAL systems in educational settings raises critical ethical and accessibility challenges. Physiological signals such as HRV and GSR are considered sensitive biometric data,[20] and their collection requires strict adherence to ethical standards and data protection regulations [96], [97]. Informed consent from learners and their guardians is essential to ensure transparency regarding data collection, use, and potential implications for learning [98]. Privacy concerns are especially salient when biometric data are continuously collected through wearable devices. Risks such as unintended surveillance, behavioral profiling, and data breaches may undermine user trust in Edtech [99]. Moreover, the ownership of physiological data often remains ambiguous, with unclear boundaries between learners, institutions, and third-party vendors [100]. In parallel, accessibility presents a practical barrier to equitable deployment. While consumer-grade wearables like smartwatches offer a scalable sensing platform, their availability remains uneven across schools and learners [101]. Socioeconomic disparities, limited infrastructure, and inconsistent device maintenance can hinder real-time PAL adoption in under-resourced environments [102], [103]. Furthermore, individual variability in skin tone, anatomy, or device usage can affect signal quality [93], raising concerns about fairness in adaptive responses. To support responsible and inclusive implementation, PAL systems should adopt privacy-by-design principles, support opt-in/opt-out participation, and promote interoperable architectures that accommodate heterogeneous hardware. Clear governance frameworks that address consent, privacy, ownership, and equitable access are essential to building trustworthy and scalable PAL ecosystems.

B. PAL MODEL EXPLAINABILITY AND TRANSPARENCY

As the proposed PAL model in Section IV relies on both rule-based and AI-driven strategies to make real-time instructional decisions based on physiological signals, the issue of explainability becomes central to their responsible deployment in educational contexts. While rule-based policies are generally perceived as inherently interpretable, this assumption may not hold—especially when policies involve complex thresholds, multiple physiological indicators, or domain-specific heuristics. Such complexity can obscure the rationale behind system adaptations, making it difficult for users to understand why specific instructional changes occur.

Similarly, AI-driven policies, including those based on decision trees or deep learning, may achieve high predictive accuracy but often operate as black-box models (i.e., models with limited interpretability) [104], [105], [106]. This opacity poses particular challenges in education, where learners and instructors must be able to understand and trust the rationale behind system-driven adaptations. A lack of transparency can make system behavior appear arbitrary or unfair, undermining user trust and acceptance [107]. Furthermore, the interpretability challenge is exacerbated by the inherent noisiness, contextual dependence, and inter-individual variability of physiological signals [94]. Therefore, system designers must prioritize not only accuracy and responsiveness, but also the clarity and traceability of decision-making across both rule-based and AI-based approaches.

Recent advances in explainable AI (XAI) offer promising tools to mitigate these concerns [108]. Post-hoc explanation techniques such as SHAP [109] and Local Interpretable Model-agnostic Explanations (LIME) [110] can attribute predictions to specific input features, allowing instructors to interpret, validate, or override system outputs [111]. Alternatively, inherently interpretable models—such as decision trees, rule-based engines, or attention-based neural architectures—can offer greater transparency while maintaining reasonable accuracy [112]. Future research should explore how such explainability techniques can be embedded into PAL systems to support human-in-the-loop adaptation and pedagogical oversight. Ensuring that adaptation decisions are not only effective but also explainable is essential to fostering trust, fairness, and accountability in AI-enhanced learning environments.

C. STANDARDIZATION IN LEARNING ANALYTICS

While traditional learning analytics has achieved some level of standardization for non-real-time data sources—such as activity logs, quiz scores, and clickstream data—the integration of real-time physiological signals remains largely unstandardized [113]. This lack of standardization poses significant barriers to interoperability, reproducibility, and scalable deployment of PAL systems. Discrepancies in sensor modalities, sampling frequencies, signal preprocessing methods, and annotation protocols hinder cross-study comparability and limit the replicability of findings in diverse educational contexts [114], [115]. Additionally, many existing datasets lack clear documentation or shared schemas, making data reuse and model transferability challenging.

To advance the field, it is essential to establish standardized data schemas for physiological signals—covering consistent

[20]HRV and GSR are particularly sensitive because they reflect involuntary autonomic responses linked to emotional and cognitive states, including stress and arousal. They may also serve as biometric identifiers, warranting stricter ethical and data protection considerations.

cognitive and affective state labels, timestamp alignment, and sensor metadata. Benchmark datasets—such as the PSI dataset [40] used in this study—can provide a common reference point for evaluating PAL systems on consistent metrics. However, future research should prioritize the development of more comprehensive datasets that incorporate richer multimodal signals, longitudinal session data, and diverse learner populations. Such datasets are crucial not only for improving the robustness and generalizability of PAL models, but also for testing and refining emerging standards in real-world settings. Aligning physiological learning analytics with existing educational data standards, such as the Experience API (xAPI) [116] and Caliper Analytics [117], may further facilitate integration into broader EdTech ecosystems. Such efforts would promote not only technical interoperability, but also cross-institutional collaboration and cumulative progress in physiology-informed learning research.

D. DATASET LIMITATIONS AND GENERALIZABILITY

Although the PSI dataset [40] serves as a useful benchmark for evaluating physiology-informed PAL systems, several structural characteristics limit its generalizability. First, the dataset provides only single-timepoint physiological recordings rather than continuous or session-long measurements. This snapshot-level structure prevents the analysis of temporal patterns such as fluctuations in cognitive load, recovery dynamics, or accumulated fatigue—factors that are central to real-world adaptive learning. To address this limitation, the present study treats PSI values as initial physiological conditions and generates multi-session, time-varying trajectories using the dynamics models described in Section III-B. However, because collecting rich, time-evolving physiological signals in authentic learning environments is inherently challenging, dynamics-based simulation should ultimately be viewed as an approximation; the validity of such simulated trajectories must be empirically verified against real longitudinal physiological measurements in future studies.

Second, as summarized in Table 1, several physiological measures (e.g., HRV and GSR) exhibit relatively narrow or shifted distributions across learners, reflecting inherent limitations of the PSI dataset. In addition, the identical value ranges reported for multiple EEG bands appear unusual and may reflect inconsistencies in annotation or preprocessing. Such issues limit the extent to which these signals represent natural physiological variability. In contrast, other variables (e.g., HR, BP, and skin temperature) show distributions that more closely align with expected physiological ranges; nevertheless, the overall diversity remains insufficient for modeling the wide spectrum of learner states encountered in authentic learning environments. Future datasets should therefore aim to capture broader and more realistic physiological variability to more accurately represent diverse learner profiles.

Third, although the PSI dataset contains eight physiological features, it does not include several modalities widely recognized in cognitive and affective computing research as informative indicators of mental and emotional states, such as pupil dilation, facial expression dynamics, and respiration variability. In our simulations, we focused on three wearable-compatible signals (HR, RR, and HRV) because these signals can be reliably captured using consumer-grade wearable devices (e.g., smartwatches), thereby supporting feasibility for real-world educational deployment. Nevertheless, expanding future datasets to incorporate additional physiological modalities would improve the robustness, redundancy, and multimodal richness of physiology-informed PAL systems.

Finally, the PSI dataset lacks behavioral learning performance metrics, including accuracy, comprehension of instructional content, retention, or task completion time, preventing empirical estimation of how physiological deviations translate into measurable learning outcomes. Consequently, this study adopts a mathematical, simulation-based framework to model the complex relationship between physiological signals and learning performance. Because establishing an exact mapping between physiological deviations and behavioral learning outcomes remains an inherently challenging problem, the mathematical formulation presented here should be viewed as a reference model that offers initial theoretical insight. Once empirical multi-session physiological–performance datasets become available, future work will extend this formulation to nonlinear or data-driven performance models that more faithfully capture real cognitive learning dynamics. Such advancements will also require large-scale empirical validation involving real learners, multi-session physiological recordings, and authentic learning tasks, which remains an essential direction for future research.

Altogether, advancing the generalizability and empirical grounding of PAL systems will require richer multimodal datasets, longitudinal physiological recordings, diverse learner populations, and realistic performance labels. Such datasets are crucial for improving the external validity of physiology-informed adaptive learning models and assessing how well they generalize to real-world educational environments.

VII. CONCLUSION

This paper presented a physiology-informed PAL framework that leverages real-time physiological signals to estimate learners' states and dynamically adjust instructional difficulty. A performance update model was introduced to capture session-wise changes in learner capability by incorporating both physiological signal deviations and accumulated fatigue. Based on this model, we proposed and evaluated a set of difficulty adaptation policies—comprising rule-based heuristics and AI-driven strategies—using a publicly available physiological dataset. Empirical results show that AI-driven strategies consistently outperformed rule-based approaches in optimizing simulated learning outcomes. This improvement reflects the ability of AI models to capture nonlinear relationships between physiological signals and learning performance. These findings highlight the potential of AI-based policies to enable more personalized and effective adaptation in physiology-informed PAL systems. Future work will focus on addressing remaining challenges for real-world deployment, including robustness, explainability, privacy considerations, and integration with broader EdTech infrastructures.

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JUNGMIN HWANG received the B.S. degree from the Division of AI Convergence, Hallym University, South Korea, in 2025, where she is currently pursuing the M.S. degree. Her research interests include artificial intelligence and machine learning (AI/ML), multimodal data processing, and adaptive learning.

CHEOLKYU SHIN (Member, IEEE) received the B.S. degree from Inha University, Incheon, South Korea, in 2007, and the M.S. and Ph.D. degrees from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2009 and 2013, respectively. From 2013 to 2023, he was with Samsung Research, Seoul, South Korea, where he contributed to the development and standardization of 3GPP LTE and NR technologies. In 2024, he joined the Division of AI Convergence, Hallym University, Chuncheon, South Korea, where he is currently an Assistant Professor. His research interests include smart mobility and positioning, non-terrestrial networks (NTN), educational technology (EdTech), and AI-driven networked systems and services.